Carlos Daniel Carrasco

University of California, San Francisco & San Francisco VA Medical Center
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Google Scholar — Personal Page

EDUCATION

Ph.D., Neuroscience, University of California, Davis 2024
Advisors: Steven J. Luck, Ph.D. & Erie D. Boorman, Ph.D.

B.S., Neuroscience, University of Nevada, Reno 2017
Advisor: Marian Berryhill, Ph.D.

PROFESSIONAL AND ACADEMIC APPOINTMENTS

Postdoctoral Scholar, University of California, San Francisco & San Francisco VA 2025–Present
Brain Imaging & EEG Lab (Mathalon Lab)

Visiting Scientist, Brigham and Women's Hospital, Mass General Brigham 2025–Present
AMP SCZ EEG Consortium

Research Data Specialist, California Office of Energy Infrastructure Safety 2024–2025

Graduate Student Researcher/Fellow, Teaching Assistant, UC Davis 2018–2024
Center for Mind & Brain (Luck Lab; Boorman Lab)

Jr. Research Specialist, University of Nevada, Reno 2018
Memory and Brain Lab (Berryhill Lab)

Student Research Assistant, University of Nevada, Reno 2015–2017
Memory and Brain Lab (Berryhill Lab); Neuroplasticity Lab (Jiang Lab)

PUBLICATIONS

[1] **Carrasco, C. D.**, Jones, K. T., & Berryhill, M. E. (2025). Older adults are impaired by distractors presented during working memory encoding. *Psychological Research, 89*, 160. https://doi.org/10.1007/s00426-025-02197-y

[2] **Carrasco, C. D.**, Simmons, A. M., Kiat, J. E., & Luck, S. J. (2025). Enhanced working memory representations for rare events. *Psychophysiology, 62*(3), e70038. https://doi.org/10.1111/psyp.70038

[3] Gowda, H. T., Kaul, N., **Carrasco, C. D.**, Battraw, M., Amer, S., Kotwal, S., Lam, S., et al. (2025). A database of upper limb surface electromyogram signals from demographically diverse individuals. *Scientific Data, 12*(1), 517. https://doi.org/10.1038/s41597-025-04825-z

[4] Zhang, G., **Carrasco, C. D.**, Winsler, K., Bahle, B., Cong, F., & Luck, S. J. (2024). Assessing the effectiveness of spatial PCA on SVM-based decoding of EEG data. *NeuroImage, 293*, 120625. https://doi.org/10.1016/j.neuroimage.2024.120625

[5] **Carrasco, C. D.**, Bahle, B., Simmons, A. M., & Luck, S. J. (2024). Using multivariate pattern analysis to increase effect sizes for event-related potential analyses. *Psychophysiology, 61*(7), e14570. https://doi.org/10.1111/psyp.14570

[6] Shires, J., **Carrasco, C. D.**, & Berryhill, M. E. (2020). No tDCS augmented working memory training benefit in undergraduates rewarded with course credit. *Brain Stimulation, 13*(6), 1524–1526. https://doi.org/10.1016/j.brs.2020.08.015

[7] Vercillo, T., **Carrasco, C. D.**, & Jiang, F. (2017). Age-related changes in sensorimotor temporal binding. *Frontiers in Human Neuroscience, 11*, 500. https://doi.org/10.3389/fnhum.2017.00500

MANUSCRIPTS IN PREPARATION

Carrasco, C. D., Light, G. A., … Mathalon, D. H. (in preparation). *Reduced P300 response with condition-specific habituation differences in clinical high-risk for psychosis: Convergent evidence from ERP amplitude and multivariate decoding.*

Carrasco, C. D., … Mathalon, D. H. (in preparation). *Deficient alpha-band event-related desynchronization during auditory oddball task performance is associated with increased risk for psychosis in clinical high risk youth.*

Carrasco, C. D. (in preparation). *Wide-coverage intracranial recordings reveal onset-locked P3b and alpha desynchronization that extend toward action.*

Wong, M.*, **Carrasco, C. D.***, … Mathalon, D. H. (in preparation). *A somatosensory oddball P300 marker of psychosis risk.*
*Co-first authors.

TEACHING EXPERIENCE

PSC 001Y, General Psychology — Teaching Assistant, UC Davis	2020, 2021, 2024
PSC 133, NeuroEconomics — Teaching Assistant, UC Davis	2021
NSC 298, Neuroimaging Journal Club — Discussion Leader, UC Davis	2018–2019
Neuroscience Boot Camp — Live EEG Demonstration	2019–2023
Built demonstration software, UC Davis	
SCI 110, First Year Experience — Discussion Leader, UNR	2015

MENTORSHIP

Neha Kaul (B.S. Cognitive Neuroscience, UC Davis, 2025)
Certified Clinical Medical Assistant (CCMA)

Saniya Kotwal (B.S. Cognitive Science, UC Davis, 2025)
Associate Product Manager, ShipStation Global

Safa Amer (B.S. Neurobiology, Physiology, and Behavior, UC Davis, 2025)
Behavior Specialist

Sebastian Lopez (B.S. Cognitive Neuroscience, UC Davis, 2024)
Ph.D. Student, University of Nevada, Reno

Samantha Fong (B.S. Cognitive Science, UC Davis, 2023)
Ph.D. Student, UC Santa Cruz

Emma Ratnaparkhi (B.S. Neurobiology, Physiology, and Behavior, UC Davis, 2022)
M.D. Student, Frank H. Netter MD School of Medicine, Quinnipiac University

Seigo Ninomiya (B.S. Cognitive Science, UC Davis, 2022)
Clinical Research Coordinator, Stanford University

Charlize Mitra (B.S. Neurobiology, Physiology, and Behavior, UC Davis, 2022)
Nursing Student, Samuel Merritt University

GRANT-SUPPORTED RESEARCH

Doctoral research (Luck Lab, UC Davis; PI: Steven J. Luck, Ph.D.)
R01EY033329 (National Eye Institute); R01MH087450 and R01MH076226 (National Institute

of Mental Health)
Funded my published EEG/ERP decoding work in NeuroImage (2024) and Psychophysiology (2024, 2025).

Postdoctoral research (Mathalon Lab, UCSF & San Francisco VA)
Mentor: Daniel H. Mathalon, Ph.D., M.D.
PREDICT-DPACC / AMP-SCZ — NIMH U24MH124629
Data Processing, Analysis, and Coordination Center
Grant PIs: Martha Shenton and René Kahn
Supports my postdoctoral EEG analyses of psychosis risk.

FELLOWSHIPS, HONORS, AND AWARDS

President's Pick Paper Award 2026
Society for Psychophysiological Research
For "Using multivariate pattern analysis to increase effect sizes for event-related potential analyses" (Carrasco et al., *Psychophysiology*, 2024).

Student Seminar Series Lead Speaker (twice selected) 2022, 2023
UC Davis Cognitive Neuroscience Area

DataBlitz Speaker — Cognitive Neuroscience Society, San Francisco, CA 2022

Graduate Student Association Travel Award — UC Davis 2021, 2023

Learning, Memory, and Plasticity (LaMP) NIH Fellow — UC Davis 2019–2020

Initiative for Maximizing Student Development (IMSD) NIH Fellow — UC Davis 2018–2019

NSF EPSCoR Undergraduate Research Opportunity Program (UROP) Award — University of Nevada, Reno 2017

Ronald E. McNair Postbaccalaureate Achievement Program Scholar — University of Nevada, Reno 2016–2017

Nevada Undergraduate Research Award — University of Nevada, Reno 2016

SERVICE AND OUTREACH

Undergraduate Speaker, Advancing Diversity in Neuroscience Research — UC Davis, 2024
Undergraduate Mentor, Advancing Diversity in Neuroscience Research — UC Davis, 2023
McNair Scholars Program Speaker & Panelist (graduate school & research paths) — UC Davis, 2021
Neuroscience Graduate Group Peer Mentor — UC Davis, 2020
NeuroFest Organizer — UC Davis, 2020 (event cancelled due to COVID-19)
Elementary School Brain Awareness Outreach — UC Davis (2018–2019); UNR (2017)

INVITED TALKS

Dashboarding and Data Science for Decision-Making. California Natural Resources Agency, Sacramento, 2024

Multivariate EEG Approaches to Working Memory and Contingency Learning. Knight Lab (Prof. Robert T. Knight), UC Berkeley, 2023

Multivariate EEG Approaches, Working Memory, and Contingency Learning. Cognitive Brain Sciences Early Career Seminar Series, UNR, 2023

Are Rare Stimuli that Produce a P3b Component Preferentially Encoded in Working Memory?. Cognitive Neuroscience Society, San Francisco, 2022

A Comparison of ERPs, SVM Decoding, and Cross Mahalanobis Distance Analysis in EEG. UCD EEG Meeting, UC Davis, 2021

Behavioral Oscillations in Working Memory. Ronald McNair Scholars Symposium, UC Berkeley, 2017

CONFERENCE POSTERS

Carrasco CD, Light GA, … Mathalon DH. (2026). *Neural Habituation Deficits in Clinical High-Risk Youth.* Society of Biological Psychiatry (SOBP); **Biological Psychiatry, 99**, S416–S417. https://doi.org/10.1016/j.biopsych.2026.03.978

Kaul N, Amer S, Kotwal S, Gowda HT, **CD Carrasco**, Miller LM. *Towards Feasible EEG-Based BCIs for Hand Gesture Control.* UC Davis Undergraduate Research Conference, 2024

Kaul N, Amer S, Kotwal S, Gowda HT, **CD Carrasco**, Miller LM. *Towards Feasible EEG-Based BCIs for Hand Gesture Control.* Center for Mind and Brain Annual Poster Day, UC Davis, 2024

Emergent Model-Based Replay as a Function of Human Credit Assignment. Society for Neuroscience, Washington, D.C., 2023

Rewards Elicit Choice Reactivation for Credit Assignment. Society for Neuroscience, San Diego, 2022

Working Memory and Influences of Selective Attention. Nevada Undergraduate Research Symposium, UNR, 2017

Age-Related Changes in Sensorimotor Temporal Binding. International Multisensory Research Forum, Vanderbilt University, 2017

Working Memory and Selective Attention. Psi Chi Annual Research Symposium, UNR, 2016

SKILLS

EEG/ERP methods: Recording, preprocessing, and analysis; EEGLAB/ERPLAB, MNE-Python, FieldTrip
Machine learning and statistics: Multivariate decoding (LDA/SVM, MVPA), spatial and temporal PCA, mixed-effects modeling
Programming and experiments: Python, R, MATLAB, Bash; PsychoPy
Computing: Linux/Unix, HPC (SLURM, SGE), cloud computing, Git/GitHub
Neuroimaging workflows: BIDS, fMRIPrep